

Tuomas Kuusisto

tuokuu99@gmail.com | (+358)503130730 | Helsinki | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

ABOUT ME

Junior Data Scientist working in R&D across machine learning, computer vision, and multimodal data. At Cense Analytics, I develop and evaluate ML models for sensor and image data using Python and TensorFlow, and work with SQL, Databricks, and data pipelines. In personal projects, I build LLM applications and AI agents using the OpenAI API, including retrieval and automated evaluation.

WORK EXPERIENCE

Cense Analytics (Junior Data Scientist)

2025 autumn ➡

- Developed CNN-based machine learning models for image data analysis.
- Developed a multimodal machine learning model combining LIBS spectroscopy data with microscope images.
- Contributed to R&D projects by evaluating new machine learning methods and presenting findings in internal team reviews.
- Executed complex data analysis and management tasks using SQL and Python within the Databricks environment.
- Preprocessed large-scale datasets using OpenCV, NumPy, and Pandas, and created visualizations using Matplotlib to support data analysis.
- Evaluated and improved existing machine learning models through statistical analysis, feature engineering, hyperparameter optimization, validation, and error analysis.

Cense Analytics (Summer Employee)

2025 summer

- Various tasks including soil sampling, company promotion at trade fairs, customer meetings, and IT support duties.

KEY COURSES & PROJECTS

Programming 1 & 2, Databases and Information Management, Software Engineering *University of Jyväskylä*

- Built a strong foundation in programming, databases, software engineering, testing, and modern development practices.

AI Portfolio Assistant

- Built and deployed an AI-powered portfolio chatbot using Python, FastAPI, the OpenAI API, and a responsive HTML, CSS, and JavaScript interface.
- Integrated CV, project README documents and portfolio as source material, enabling answers with source citations and access to the original documents.
- Compared full-context prompting, local retrieval-augmented generation (RAG), and OpenAI File Search through structured evaluations of answer quality, response time, and token usage.
- Implemented structured-output validation, citation checks, request limits, and handling of unsupported or unrelated questions.
- Project links: [Live demo](#) | [GitHub repository](#) (Demo may take a minute to load due to render's hosting tier)

Data Dashboard Project: Air Strikes on Ukraine Analysis

- Built and published an MVP analytics dashboard in Streamlit for monitoring Russian air strikes against Ukraine using Python, SQL, and DuckDB.
- Designed and maintained data transformation workflows and analytical data marts to support interactive filtering, time-based analysis, and regional strike monitoring.
- Executed exploratory and comparative analysis on weapon systems, strikes, and geographic targeting patterns using SQL and Python.
- Developed map-based and time-series visualizations to support analysis of weapon usage, air defense outcomes, and operational trends.
- Project links: [Streamlit dashboard](#) | [GitHub repository](#)

REFERENCES

Dr. Paavo Kajovirta CTO at Cense Analytics

- Contact information upon request

EDUCATION

University of Jyväskylä *Bachelor's and Master's Programme in Computer Science*

Present

- B.Sc. in Computer Science - expected October 2026
- Minor studies in Statistics and Mathematics

VOLUNTEER EXPERIENCE

Sports Coordinator *Linkki ry*

- Organized sports trials and weekly exercise sessions for student organization members.

Event Team Member *Linkki ry*

- Planned and executed student events collaboratively.

SKILLS & INTERESTS

- Programming Languages & Technologies: Java, C#, Python, TensorFlow, SQL, Pandas, Numpy, OpenCV, JavaScript, HTML, CSS, React, Git, Robot Framework, Pytest, Databricks, Streamlit, R
- Interests: Machine learning, statistics, data, technological innovations, basketball